

Recurrence Relations: In-Class Practice (30 points, 10 each)

Solve each recurrence using the Master Theorem below. Show your work.

The Master Theorem (Roughgarden)

Let $a \geq 1$, $b \geq 2$, and $d \geq 0$ be constants, and suppose $T(n)$ is defined on the nonnegative integers by the recurrence

$$T(n) \leq a \cdot T(n/b) + O(n^d), \quad T(1) \leq c,$$

where n/b means either $\lceil n/b \rceil$ or $\lfloor n/b \rfloor$, and c is a constant. Then

$$T(n) = \begin{cases} O(n^d \log n) & \text{if } a = b^d \quad [\text{Case 1}] \\ O(n^d) & \text{if } a < b^d \quad [\text{Case 2}] \\ O(n^{\log_b a}) & \text{if } a > b^d \quad [\text{Case 3}] \end{cases}$$

How to use it: write your recurrence as $T(n) = aT(n/b) + f(n)$, identify a , b , and d from $f(n) = O(n^d)$, then compare a to b^d to pick the matching case above.

Worked Example

Here is a fully worked example showing how to apply the theorem. Use this as a model for how to write up your own solutions to the problems that follow.

Example (Binary Search). Solve $T(n) = T(n/2) + 1$.

Identify a , b , $f(n)$: $a = 1$, $b = 2$, $f(n) = 1 = O(n^0)$, so $d = 0$.

Compare a to b^d : $b^d = 2^0 = 1$. Since $a = 1 = b^d$, we are in **Case 1**.

Apply the theorem: Case 1 gives $T(n) = O(n^d \log n) = O(n^0 \log n) = O(\log n)$.

Final Answer: $T(n) = \Theta(\log n)$

(Case 1: $a = b^d$)

Problem 1. Solve the following recurrence using the Master Theorem.

$$T(n) = 2T(n/2) + 3n$$

Work:

Problem 2. Solve the following recurrence using the Master Theorem.

$$T(n) = 3T(n/4) + \sqrt{n}$$

Work:

Problem 3. Solve the following recurrence using the Master Theorem.

$$T(n) = 7T(n/2) + \Theta(n^3)$$

Work: